Linear Equations Solver

A C++ Object-Oriented System for Parsing, Manipulating & Solving Linear Systems

$$Ax = b$$

Gaussian Elimination | Cramer's Rule | Row Operations | C++ OOP

Project Overview

Purpose

Build a C++ program that reads, stores, and manipulates systems of linear equations – supporting row operations and two solving strategies: Gaussian Elimination and Cramer's Rule.

Scope

Handles n-variable systems entered as text. Supports parsing, printing, column extraction, addition, subtraction, substitution, and matrix-based solutions.

Design

Two-class OOP architecture: Equations holds the data model (coefficients + constant), LinearSys inherits it and provides all operations and solvers.

Interface

Interactive command-line loop. Accepts: num_vars, equation, column, add, subtract, substitute, D, D xN, D_value, solve, quit.

Class Architecture

Equations

Fields

vector<float> coefficients
float con

Constructor

Equations(int vars = 0)
resizes coefficient vector
initialises con = 0

inherits

LinearSys : Equations

Fields

vector<Equations> equations
int numVars

Methods

addEquation() num_vars() printEquation()
column() add() subtract()
substitute() D() D_var() D_value()
solve() — Gaussian & Cramer's

Command Reference

Command	Description	Example
`num_vars`	Print number of variables in the system	`num_vars`
`equation i`	Print equation i with all its coefficients	`equation 2`
`column xi`	Print all coefficients for variable xi	`column x1`
`add a b`	Print the sum of equations a and b	`add 1 2`
`subtract a b`	Print the difference of equations a and b	`subtract 2 3`
`substitute xi a b`	Eliminate xi from equation a using equation b	`substitute x2 1 3`
`D`	Print the Cramer's coefficient matrix	`D`
`D xi`	Cramer's matrix with constants replacing column xi	`D x2`
`D_value`	Print the determinant of the coefficient matrix	`D_value`
`solve`	Solve the system via Gaussian Elimination	`solve`

Equation Parsing — addEquation()

1. **Read input string**
   e.g. "3x1+2x2+1x3=12"

2. **Split on '='**
   left = "3x1+2x2+1x3" right = "12"

3. **atof(right)**
   con = 12.0

4. **Tokenise left on '+'**
   tokens: "3x1", "2x2", "1x3"

5. **Per token: find 'x'**
   coef = atof(before x), index = atoi(after x)

6. **Store in vector**
   coefficients[index-1] = coef

addEquation() — Core Logic

```
int pos = input.find('=');
eq.con = atof(right);

while getline(ss, term, '+') {
  xPos = term.find('x');
  coef  = atof(term[0..xPos]);
  index = atoi(term[xPos+1]);
  coefficients[index-1] = coef;
}
equations.push_back(eq);
```

Row Operations

add(a, b)

```
eq_a[j] + eq_b[j]
```

Adds coefficients and constants from equations a and b element-by-element and prints the resulting equation. Does not modify stored equations.

```
add 1 2  →  3x1+4x2+3x3=26
```

subtract(a, b)

```
eq_a[j] - eq_b[j]
```

Subtracts equation b from equation a term-by-term. Useful for manual Gaussian-style elimination steps.

```
subtract 2 1  →  -1x1+2x2+1x3=8
```

substitute(xV, a, b)

```
ratio = eq_a[xV] / eq_b[xV]
```

Eliminates variable xV from equation a using equation b as the source. Computes ratio then applies: coef[j] = coef_a[j] − ratio × coef_b[j]. Skips zero terms.

```
substitute x2 1 2  →
1.667x1+0.333x3=3.33
```

Gaussian Elimination — solve()

Phase 1 — Build Matrix

- Copy coefficients into n×(n+1) augmented matrix
- Last column stores the constant (con) of each equation

Phase 2 — Forward Elimination

- For each pivot row i: check matrix[i][i] ≠ 0
- For each row j below: ratio = matrix[j][i] / matrix[i][i]
- Subtract ratio × pivot row from row j
- Produces upper-triangular form

Phase 3 — Back Substitution

- Start from last equation (1 unknown)
- Solve upward: solution[i] = (rhs − known terms) / diagonal
- Print x1= x2= x3= etc

Pseudocode

```
// forward elimination
for i in 0..n:
  if matrix[i][i] == 0 → No Solution
  for j in i+1..n:
    ratio = matrix[j][i] / matrix[i][i]
    matrix[j][k] -= ratio * matrix[i][k]

// back substitution
for i in n-1..0:
  sol[i] = matrix[i][numVars]
  for j in i+1..n:
    sol[i] -= matrix[i][j] * sol[j]
  sol[i] /= matrix[i][i]
```

Cramer's Rule — D(), D_var(), D_value()

Cramer's Rule states: xi = det(Ai) / det(A)

where A is the coefficient matrix and Ai replaces column i with the constants vector.

D()

Prints the n×n coefficient matrix surrounded by | … | pipes.
All rows, all variable columns, no constants.

```
D
```

D_var(xi)

Replaces column xi with the constants (con) vector — the Cramer's matrix for solving for variable xi.

```
D x2
```

D_value()

Computes det(A) via recursive cofactor expansion and prints the result. Used to check solvability.

```
D_value  →  D = 13
```

Sample Session

linear_solver — bash

```
$ ./linear
Enter number of equations: 3
Enter equations:
2x1+1x2+1x3=9
1x1+3x2+2x3=17
1x1+2x2+4x3=20

> D_value
D = 13

> substitute x2 1 2
1.66667x1+0.333333x3=3.33333

> solve
x1=1
x2=2
x3=4
```

D_value

det(A) = 13 ≠ 0, so a unique solution exists. System is solvable.

substitute x2 1 2

Eliminates x2 from eq 1 using eq 2 as the source. x2 coefficient becomes 0.

solve

Gaussian Elimination finds x1=1, x2=2, x3=4. Verify: 2(1)+1(2)+1(4)=8≠9… check!

Summary

Linear Equations Solver

- OOP design — Equations data model + LinearSys operations engine cleanly separated
- Row operations: add, subtract, substitute for interactive equation manipulation
- Gaussian Elimination for fast algebraic solution of n-variable systems
- Cramer's Rule support: D(), D_var(), D_value() for determinant-based solving
- Extensible architecture ready for pivoting, file I/O, and additional solvers

Built with C++ · Object-Oriented Design · Gaussian Elimination · Cramer's Rule